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Practical 08

Aim: Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Name:

USN:

Semester / Year:

4th SEM / 2nd YEAR

Academic Session:

2025-26

Date of Performance:

Date of Submission:

❖ CODE :

```
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ nano lru.c
```

```
GNU nano 8.7  
#include <stdio.h>  
  
int main() {  
    int pages[50], frames[10];  
    int n, f, i, j, pos, faults = 0, hits = 0;  
    int least[10], count = 0;  
  
    printf("Enter number of pages: ");  
    scanf("%d", &n);  
  
    printf("Enter page reference string:\n");  
    for(i = 0; i < n; i++)  
        scanf("%d", &pages[i]);  
  
    printf("Enter number of frames: ");  
    scanf("%d", &f);  
  
    for(i = 0; i < f; i++)  
        frames[i] = -1;  
  
    for(i = 0; i < n; i++) {  
        int found = 0;  
  
        for(j = 0; j < f; j++) {  
            if(frames[j] == pages[i]) {  
                count++;  
                hits++;  
                found = 1;  
                least[j] = count;  
                break;  
            }  
        }  
  
        if(!found) {  
            pos = 0;  
            for(j = 1; j < f; j++)  
                if(least[j] < least[pos])  
                    pos = j;  
  
            frames[pos] = pages[i];  
            count++;  
            least[pos] = count;  
            faults++;  
        }  
    }  
  
    printf("\nTotal Page Hits = %d", hits);  
    printf("\nTotal Page Faults = %d", faults);  
  
    float hit_per = (float)hits/n * 100;  
    float fault_per = (float)faults/n * 100;  
  
    printf("\nHit Percentage = %.2f%%", hit_per);  
    printf("\nFault Percentage = %.2f%%", fault_per);  
  
    return 0;  
}
```

❖ OUTPUT :

```
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ gcc lru.c -o lru  
  
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ ./lru  
Enter number of pages: 10  
Enter page reference string:  
3 4 5 7 8 4 1 6 4 9  
Enter number of frames: 3  
  
Total Page Hits = 1  
Total Page Faults = 9  
Hit Percentage = 10.00%  
Fault Percentage = 90.00%  
Tejaswini@LAPTOP-COS64E96 MSYS ~
```